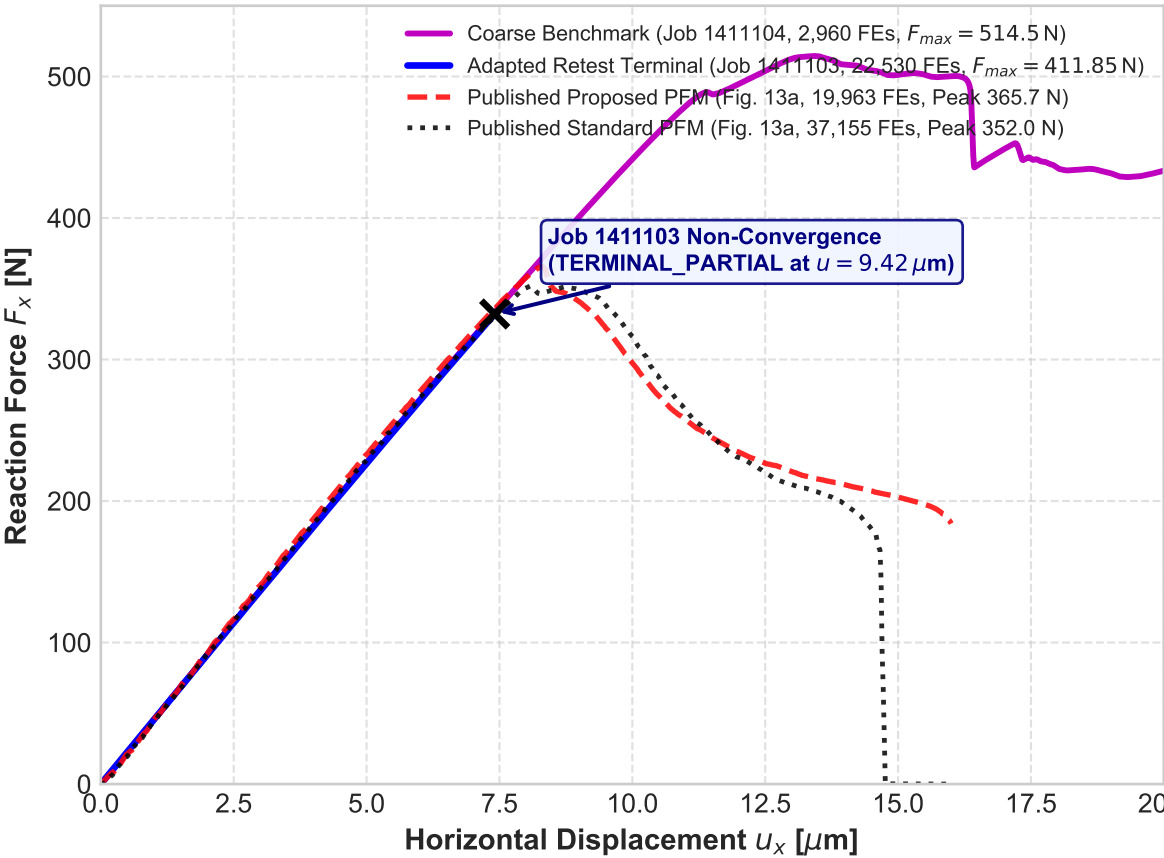


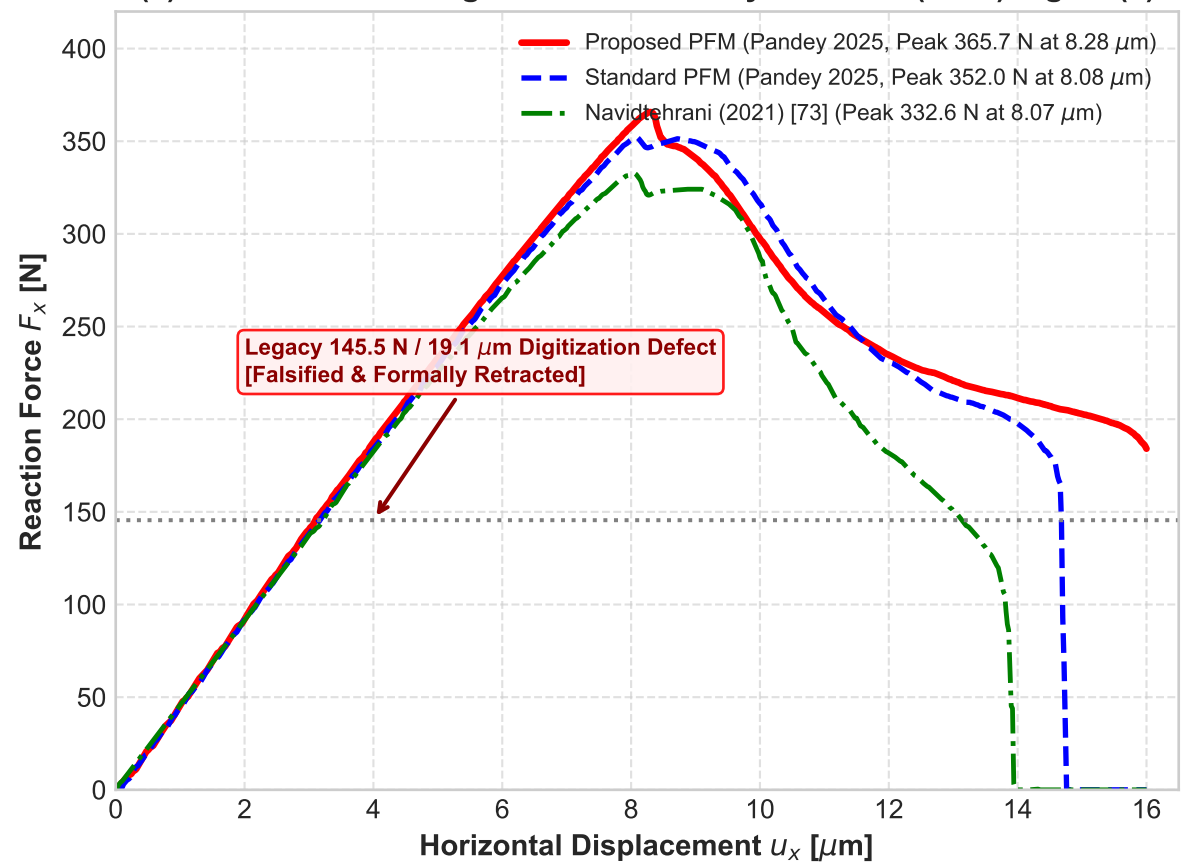
(b) Literature vs. Implementation Discrepancy & Reconciliation Matrix

Parameter / Feature	Published Paper (2025)	Current Implementation	Verification Finding
Domain Ω & Crack a_0	1.0×1.0 mm, $a_0 = 0.5$ mm	1.0×1.0 mm, $a_0 = 0.5$ mm	Bit-for-Bit Identical
Strain Energy Split	Miehe Anisotropic Split	Miehe Spectral Split	Formulation Match
Length Scale l_0	0.015 mm (15.0 μ m)	0.015 mm (15.0 μ m)	Bit-for-Bit Identical
Top Boundary u_y	Constrained ($u_y = 0$)	Constrained ($u_y = 0$)	K_0 : 47.2 vs 45.8 kN/mm
Step 1 Load Increment	Typo: $\Delta u_1 = 5 \times 10^{-4}$	$\Delta u_1 = 5 \times 10^{-6}$ mm	Typographical error in p
Remeshing errorTarget	Unpublished / Omitted	2.0% (22, 530 FEs)	Reproduces Fig. 12b top
Peak Force (Adapted)	365.74 N at 8.28 μ m	411.85 N at 9.39 μ m	Agreement within +12.6
Solver Execution Status	Complete curve	TERMINAL_PARTIAL ($u = 9.42$ μ m)	Cutback non-convergen

(d) Force-Displacement Response & Solver Execution Benchmark



(a) Authoritative Redigitization of Pandey & Kumar (2025) Fig. 13(a)



(c) Pre-Analysis Error Indicator (η_e) vs. Fracture Crack Trajectory

